

GML_03_02-rating.R

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```
# mar 2024
# r script for simulation of BINARY RESPONSE MODELS
# written by Frank Lehrbass (->lehrbass.de) for teaching purposes only
# students are solely responsible for whatever they do with this coding
# this R coding is WITHOUT ANY WARRANTY

#data from example in book M. Verbeek (2012), A Guide to Modern Econometrics,
p 223

# Step 1

rm(list=ls(all=TRUE))

# Load packages

library(logistf)
library(GGally)

## Lade nötiges Paket: ggplot2

## Warning: Paket 'ggplot2' wurde unter R Version 4.3.3 erstellt

## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg      ggplot2

library(pscl)

## Classes and Methods for R developed in the
## Political Science Computational Laboratory
## Department of Political Science
## Stanford University
## Simon Jackman
## hurdle and zeroinfl functions by Achim Zeileis

library(lmtest)      # Tests für (generalisierte) lineare Modelle

## Warning: Paket 'lmtest' wurde unter R Version 4.3.3 erstellt

## Lade nötiges Paket: zoo

##
## Attache Paket: 'zoo'
```

```
## Die folgenden Objekte sind maskiert von 'package:base':
##
##      as.Date, as.Date.numeric

library(broom)      # Ausgabe und Formatierung verschiedener Objekte
library(pscl)       #ord Logit
library(MASS)

dfrm_clean <- read.table("GML_03_02_rating.csv", header=TRUE,
sep=";", dec=",")

dfrm = as.data.frame(dfrm_clean)
dfrm[1:5,]

##      BOOKLEV      EBIT INVGRADE LOGSALES      MARKLEV RATING      RETA
## 1 0.2566776 0.057609940      0 6.316473 0.21276317      2 -0.1079751
## 2 0.4982200 0.003254789      0 9.938469 0.78343530      2 -0.1068656
## 3 0.2276335 0.166817918      1 10.011795 0.09852892      6 0.3826220
## 4 0.4676779 -0.029462988      0 8.430120 0.64484390      2 -0.1430880
## 5 0.0000000 0.034412265      0 7.291656 0.00000000      3 0.2852240
##
##      WKA
## 1 0.22902174
## 2 -0.07309713
## 3 0.13625087
## 4 0.03182491
## 5 0.65084680

#remove incomplete rows
dfrm_clean = na.omit(dfrm)

#cross check with txt file descr, also page 223 in book
summary(dfrm_clean)

##      BOOKLEV      EBIT      INVGRADE      LOGSALES
## Min.      :0.0000  Min.      :-0.38417  Min.      :0.0000  Min.      : 1.100
## 1st Qu.:0.1698  1st Qu.: 0.05120  1st Qu.:0.0000  1st Qu.: 6.970
## Median :0.2640  Median : 0.09044  Median :0.0000  Median : 7.884
## Mean    :0.2932  Mean    : 0.09389  Mean    :0.4723  Mean    : 7.996
## 3rd Qu.:0.3877  3rd Qu.: 0.13599  3rd Qu.:1.0000  3rd Qu.: 8.950
## Max.    :0.9992  Max.    : 0.65151  Max.    :1.0000  Max.    :12.701
##      MARKLEV      RATING      RETA      WKA
## Min.      :0.0000  Min.      :1.000  Min.      :-0.99589  Min.      :-0.41208
## 1st Qu.:0.1091  1st Qu.:3.000  1st Qu.: 0.01108  1st Qu.: 0.02905
## Median :0.2112  Median :3.000  Median : 0.18048  Median : 0.12284
## Mean    :0.2547  Mean    :3.499  Mean    : 0.15699  Mean    : 0.14041
## 3rd Qu.:0.3477  3rd Qu.:4.000  3rd Qu.: 0.35084  3rd Qu.: 0.23462
## Max.    :0.9649  Max.    :7.000  Max.    : 0.97992  Max.    : 0.74802

#to be explained
rating <- dfrm_clean$RATING #the higher the better, <3 = non IG
IG<-dfrm_clean$INVGRADE
```

```

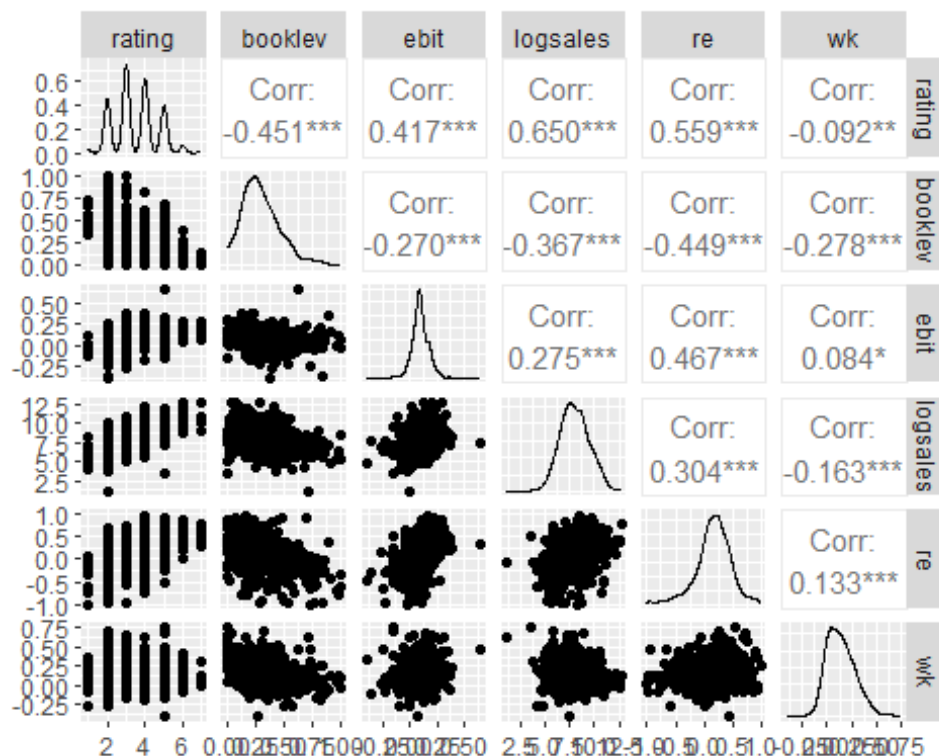
#the regressors
booklev <- dfrm_clean$BOOKLEV #for Leverage
ebit <- dfrm_clean$EBIT #current profitability, already divided by ta!
logsales <- dfrm_clean$LOGSALES #for firm size
wk <- dfrm_clean$WKA #proxies liquidity WC
re <- dfrm_clean$RETA #retained earnings = past profitability

X = cbind(booklev,ebit,logsales,re,wk)
Data = cbind(IG,X)
X = as.data.frame(X)

x11()#again no simple rules visible
ggpairs(cbind(rating,X))

## Warning in geom_point(): All aesthetics have length 1, but the data has 36
rows.
## i Did you mean to use `annotate()`?

```



```

#end data mining ... start binary choice modelling

# Step 2

#NOW LOGIT model as in book-----

#use logistf for binary logit way 1
fit1 <- logistf(IG~booklev+ebit+logsales+re+wk,pl=FALSE, method =
"Standard ML", firth = F)
fit1

```

```

## logistf(formula = IG ~ booklev + ebit + logsales + re + wk, pl = FALSE,
##       firth = F, method = "Standard ML")
## Model fitted by Standard ML
## Confidence intervals and p-values by Wald
##
## Coefficients:
## (Intercept)      booklev          ebit      logsales          re          wk
##   -8.214322   -4.427267    4.354736    1.081593    4.116108   -4.012493
##
## Likelihood ratio test=591.7964 on 5 df, p=0, n=921

  exp(coef(fit1))

## (Intercept)      booklev          ebit      logsales          re
wk
## 2.707481e-04 1.194710e-02 7.784627e+01 2.949374e+00 6.132014e+01
1.808825e-02

#note that LR test amounts to 585, compares Loglik of model,
#where all paras except intercept are set to zero (Verbeek, 212),
#to the estimated model. The p value (p=0) shows that H0 "both equal"
#can be discarded with near certainty
fit1$loglik

##      full      null
## -341.0776 -636.9758

#logit estimation way 2
fit2 = glm(IG~booklev+ebit+logsales+re+wk, family = "binomial")

#LR either way
fit1

## logistf(formula = IG ~ booklev + ebit + logsales + re + wk, pl = FALSE,
##       firth = F, method = "Standard ML")
## Model fitted by Standard ML
## Confidence intervals and p-values by Wald
##
## Coefficients:
## (Intercept)      booklev          ebit      logsales          re          wk
##   -8.214322   -4.427267    4.354736    1.081593    4.116108   -4.012493
##
## Likelihood ratio test=591.7964 on 5 df, p=0, n=921

# Anzahl der Freiheitsgrade = Anzahl Koeffizienten minus Intercept
df <- length(coef(fit2)) - 1
# Chi2-Test
pchisq(fit2$null.deviance - fit2$deviance, df, lower.tail = FALSE)

## [1] 1.197621e-125

## R-Code 4.11 ----
lrtest(fit2)

```

```

## Likelihood ratio test
##
## Model 1: IG ~ booklev + ebit + logsales + re + wk
## Model 2: IG ~ 1
##   #Df  LogLik Df Chisq Pr(>Chisq)
## 1    6 -341.08
## 2    1 -636.98 -5 591.8  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

#compare
coef(fit1)

## (Intercept)      booklev          ebit      logsales          re          wk
##   -8.214322   -4.427267    4.354736    1.081593    4.116108   -4.012493

coef(fit2)

## (Intercept)      booklev          ebit      logsales          re          wk
##   -8.214322   -4.427267    4.354736    1.081593    4.116108   -4.012493

#nice
summary(fit2)

##
## Call:
## glm(formula = IG ~ booklev + ebit + logsales + re + wk, family =
"binomial")
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -8.21432    0.86685  -9.476  < 2e-16 ***
## booklev     -4.42727    0.77142  -5.739 9.52e-09 ***
## ebit         4.35474    1.43992   3.024 0.00249 **
## logsales     1.08159    0.09568  11.304 < 2e-16 ***
## re           4.11611    0.48851   8.426  < 2e-16 ***
## wk          -4.01249    0.74791  -5.365 8.10e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1273.95  on 920  degrees of freedom
## Residual deviance:  682.16  on 915  degrees of freedom
## AIC: 694.16
##
## Number of Fisher Scoring iterations: 6

##McFaddens Pseudo R2 just for info but no added val
pR2(fit2)[4]

## fitting null model for pseudo-r2

```

```

## McFadden
## 0.464536

# Step 3

exp(coef(fit2))

## (Intercept)      booklev      ebit      logsales      re
wk
## 2.707481e-04 1.194710e-02 7.784627e+01 2.949374e+00 6.132014e+01
1.808825e-02

# Residuen auslesen und Deltas berechnen, use this if u want to proceed
further w diags
DevResid <- resid(fit2)
PearsonResid <- resid(fit2, type = "pearson")
hats <- hatvalues(fit2)
DeltaChiSq <- PearsonResid^2 / (1 - hats)
DeltaBetas <- PearsonResid^2 * hats / (1 - hats)^2
ProbDach <- fitted(fit2)
# Plots anlegen
#g1 <- gf_point(hats ~ ProbDach)
#g2 <- gf_point(DeltaChiSq ~ ProbDach)
#g3 <- gf_point(DeltaBetas ~ ProbDach)
# in einer Zeile ausgegeben
#grid.arrange(g1, g2, g3, nrow = 1)

## R-Code 4.24 ----
# Identifikation
which(DeltaChiSq > 300)

## 362
## 362

## R-Code 4.28 ----
dfbetas(fit2)[which(DeltaChiSq > 300), ]

## (Intercept)      booklev      ebit      logsales      re
wk
## 0.084328984 -0.007387630 -0.098585086 -0.084371559 0.009427581
0.078755525

augment(fit2, type.predict = "response")

## # A tibble: 921 × 12
##      IG booklev      ebit logsales      re      wk .fitted .resid      .hat
. sigma
##   <int>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
<dbl>
## 1     0 0.257   0.0576     6.32 -0.108   0.229   0.0258 -0.229 0.00196
0.864
## 2     0 0.498   0.00325     9.94 -0.107  -0.0731  0.549  -1.26 0.0199
0.863

```

```

## 3      1  0.228  0.167      10.0  0.383  0.136  0.966  0.261 0.00217
0.864
## 4      0  0.468 -0.0295      8.43 -0.143  0.0318  0.118 -0.501 0.00690
0.864
## 5      0  0      0.0344      7.29  0.285  0.651  0.166 -0.602 0.0210
0.864
## 6      0  0.188  0.0410      8.67  0.0874  0.238  0.479 -1.14  0.00679
0.863
## 7      0  0.317 -0.0210      6.76 -0.0788  0.414  0.0123 -0.157 0.00165
0.864
## 8      1  0.245  0.105      9.00  0.469  0.0453  0.934  0.370 0.00307
0.864
## 9      0  0.285  0.0528      8.00  0.0385  0.0988  0.302 -0.849 0.00354
0.863
## 10     1  0.0543  0.146      8.17  0.561  0.284  0.899  0.461 0.00477
0.864
## # i 911 more rows
## # i 2 more variables: .cooksd <dbl>, .std.resid <dbl>

dfbetas(fit2)[which(DeltaChiSq > 300), ]

## (Intercept)      booklev      ebit      logsales      re
wk
## 0.084328984 -0.007387630 -0.098585086 -0.084371559 0.009427581
0.078755525

# Step 4

Y_hat = (fit1$predict>0.5)#cut off at 0.5 arbitrarily chosen
Y_real = (IG>0)

##Simple Hit rate SHR
d=Y_real
d_hat = Y_hat
tt = table(d, d_hat) #, exclude = NULL)
print(tt)

##      d_hat
## d      FALSE TRUE
## FALSE   408   78
## TRUE    77   358

ttpc = prop.table(tt, 1)#percenatge correct per category
print(ttpc)

##      d_hat
## d      FALSE TRUE
## FALSE 0.8395062 0.1604938
## TRUE  0.1770115 0.8229885

# total percent correct
sum(diag(prop.table(tt)))

```

```
## [1] 0.8317047

#hence a bit better than linear discriminant analysis
#BUT only here statistical evaluation possible!

# Step 5

#forecast Ig for a new "mean type" firm, BUT with double Leverage-----
-
newfirm = c(2*mean(booklev),mean(ebit),mean(logsales),mean(re),mean(wk))
#rep(0.5,5)
newlinpred = newfirm %*% coef(fit1)[2:6]+coef(fit1)[1]
levP = 1 /(1+exp(-newlinpred))#Less typing than P/(1-P)
levP

##           [,1]
## [1,] 0.158353

#now create the vector of a mean firm, except double the ebit and size,
#and spot its higher chance to get an IG!!!
newfirm = c(mean(booklev),2*mean(ebit),2*mean(logsales),mean(re),mean(wk))
newlinpred = newfirm %*% coef(fit1)[2:6]+coef(fit1)[1]
newP = exp(newlinpred)/(1+exp(newlinpred))#old style, watch missing -
newP

##           [,1]
## [1,] 0.9998308

# Step 6

#NOW ORDERED LOGIT model as in book-----

RatingFactor = as.factor(dfrm_clean$RATING)
#Ordered Logit Model
m <- polr(RatingFactor ~ booklev+ebit+logsales+re+wk, data = dfrm_clean,
Hess=TRUE)
summary(m)

## Call:
## polr(formula = RatingFactor ~ booklev + ebit + logsales + re +
##      wk, data = dfrm_clean, Hess = TRUE)
##
## Coefficients:
##              Value Std. Error t value
## booklev    -2.7517     0.47730  -5.765
## ebit         4.7313     0.94476   5.008
## logsales    0.9406     0.05884  15.984
## re          3.5600     0.30229  11.777
## wk         -2.5802     0.48263  -5.346
##
## Intercepts:
##              Value Std. Error t value
## 1|2 -0.3696     0.6332   -0.5837
```



```
## 2|3 4.8805 0.5208 9.3709
## 3|4 7.6259 0.5512 13.8361
## 4|5 9.8849 0.5916 16.7078
## 5|6 12.8829 0.6734 19.1315
## 6|7 14.7825 0.7841 18.8527
```

```
##
## Residual Deviance: 1930.614
## AIC: 1952.614
```

#Likelihood-Ratio-Test

```
lrtest(m)
```

```
## Likelihood ratio test
```

```
##
```

```
## Model 1: RatingFactor ~ booklev + ebit + logsales + re + wk
```

```
## Model 2: RatingFactor ~ 1
```

```
## #Df LogLik Df Chisq Pr(>Chisq)
```

```
## 1 11 -965.31
```

```
## 2 6 -1396.74 -5 862.87 < 2.2e-16 ***
```

```
## ---
```

```
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

##McFaddens Pseudo R²

```
pR2(m)[4]
```

```
## fitting null model for pseudo-r2
```

```
## McFadden
```

```
## 0.3088874
```

#this is Table 7.5 in Verbeek, feel good that coeffs are similar to logit model

#chi sq statistic is same = 862

#the intercepts are the borders for eta (n) in slides, i.e. if x is smaller than -0.36

#the corresponding firms gets worst rating = 1, best = 7 if eta is above 14.78

##Koeffiziententabelle

```
(ctable <- coef(summary(m)))
```

```
##          Value Std. Error    t value
## booklev -2.7516633 0.47730353 -5.7650177
## ebit     4.7313154 0.94475869  5.0079617
## logsales 0.9405506 0.05884227 15.9842673
## re       3.5599921 0.30229405 11.7765867
## wk      -2.5801608 0.48263239 -5.3460166
## 1|2      -0.3696356 0.63324593 -0.5837157
## 2|3       4.8805455 0.52081724  9.3709369
## 3|4       7.6258970 0.55115922 13.8361052
## 4|5       9.8849376 0.59163683 16.7077795
## 5|6      12.8828578 0.67338557 19.1314730
## 6|7      14.7825480 0.78410769 18.8527011
```

##p-Werte berechnen

```
p <- pnorm(abs(ctable[, "t value"]), lower.tail = FALSE) * 2
```

##Kombinierter Output

```
(ctable <- cbind(ctable, "p value" = p))
```

```
##              Value Std. Error    t value    p value
## booklev    -2.7516633 0.47730353 -5.7650177 8.164928e-09
## ebit        4.7313154 0.94475869  5.0079617 5.500947e-07
## logsales    0.9405506 0.05884227 15.9842673 1.644895e-57
## re          3.5599921 0.30229405 11.7765867 5.153813e-32
## wk         -2.5801608 0.48263239 -5.3460166 8.991097e-08
## 1|2        -0.3696356 0.63324593 -0.5837157 5.594116e-01
## 2|3         4.8805455 0.52081724  9.3709369 7.189148e-21
## 3|4         7.6258970 0.55115922 13.8361052 1.543459e-43
## 4|5         9.8849376 0.59163683 16.7077795 1.150402e-62
## 5|6        12.8828578 0.67338557 19.1314730 1.381160e-81
## 6|7        14.7825480 0.78410769 18.8527011 2.792149e-79
```

#Konfidenzintervalle berechnen

```
(ci <- confint(m))
```

```
## Es wird auf das Profilieren gewartet ...
```

```
##              2.5 %    97.5 %
## booklev    -3.6941494 -1.822027
## ebit        2.8875843  6.593054
## logsales    0.8266388  1.057410
## re          2.9750344  4.160875
## wk         -3.5311110 -1.638153
```

#hier assuming asymptotic normality!

#feeld again good that signs are noot changing from left to right border of conf int

##Odd-Ratios

```
exp(coef(m))
```

```
##      booklev      ebit    logsales      re      wk
## 0.06382162 113.44468414  2.56139127 35.16291947 0.07576182
```

#chk, k?nnnte man auswerten-)

```
head(cbind(predict(m), RatingFactor))
```

```
##      RatingFactor
## [1,] 2           2
## [2,] 4           2
## [3,] 5           6
## [4,] 3           2
## [5,] 3           3
## [6,] 3           2
```

#end